

HONEYWELL PROJECT REPORT

AEGIS-SOC

AI-Powered Behavioral Anomaly Detection & Intelligent SIEM Correlation Platform

Link: <https://github.com/swanandagupta/AEGIS-SOC>

A consolidated engineering & architecture report covering platform motivation, detection pipeline, hybrid AI engines, correlation logic, backend/frontend architecture, deployment, and future roadmap.

Attribute	Description
Domain	Cybersecurity — Security Operations Center (SOC) Automation
Core Approach	Hybrid AI: Deep Sequence Model + Statistical Outlier Detection + Rule Engine
Backend	Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, WebSockets, Docker
Frontend	React, TypeScript, Tailwind CSS, Recharts
Key Outcome	Explainable, risk-scored, correlated security incidents in real time

1. Platform Overview & Motivation

AEGIS-SOC is an intelligent Security Operations Center (SOC) platform designed to detect, analyze, and respond to modern cyber threats using artificial intelligence, behavioral analytics, and Security Information and Event Management (SIEM) correlation techniques. It is built around a simple premise: the attacks that hurt organizations most today rarely trip a known signature.

Traditional cybersecurity monitoring systems primarily rely on predefined signatures and static correlation rules to identify attacks. While effective against known threats, these systems struggle to detect sophisticated attacks such as insider threats, credential stuffing, lateral movement, and other behavioral anomalies that do not match existing signatures. They also generate a large number of false positives, leading to alert fatigue among security analysts and increasing the likelihood that genuine threats are overlooked.

AEGIS-SOC addresses these limitations by combining multiple detection methodologies into a unified platform capable of learning normal user behavior, identifying deviations in real time, correlating suspicious events across different systems, and providing explainable security insights through an interactive SOC dashboard.

2. Log Ingestion & Normalization

The platform begins by ingesting security logs from multiple heterogeneous sources, including Linux authentication logs, Windows event logs, VPN gateways, SSH servers, firewall appliances, cloud infrastructure, and other enterprise systems.

Since each log source follows a different format, the first stage of the pipeline performs log normalization, converting every incoming log into a standardized schema containing fields such as timestamp, username, source IP address, destination IP address, service name, event type, and message content. Standardizing logs at the ingestion stage ensures that all downstream components can process data consistently regardless of the original source format.

3. Parsing, Template Mining & Event Abstraction

After normalization, the logs are processed through a parsing and event abstraction pipeline. Instead of treating every log as a unique event, AEGIS-SOC performs template mining to identify recurring log patterns and remove dynamic values such as IP addresses, timestamps, and process IDs. This abstraction significantly reduces data complexity while preserving the semantic meaning of each event.

The parsed logs are then categorized into predefined security event classes such as authentication success, authentication failure, invalid user attempts, privilege escalation, session termination, and system events. By converting raw logs into structured event categories, the platform creates a compact representation that is easier for machine learning models and rule-based engines to analyze.

4. Feature Engineering

Once events have been standardized, the platform performs extensive feature engineering to transform raw security events into numerical representations suitable for anomaly detection. A sliding event window is maintained for every user or session, allowing the system to capture sequential behavior over time rather than analyzing events individually.

Various statistical features are extracted, including total login attempts, failed authentication counts, login success ratios, authentication frequency, privilege escalation attempts, event distributions, and temporal behavior patterns. These features provide valuable contextual information that enables the system to distinguish between legitimate user activity and potentially malicious behavior.

Figure 1 — End-to-End Detection & Correlation Pipeline



End-to-end flow from raw log ingestion through hybrid detection, fusion, correlation, and dashboard delivery.

5. Hybrid Anomaly Detection Engine

The core intelligence of AEGIS-SOC lies in its hybrid anomaly detection engine, which combines three complementary detection approaches operating simultaneously so that behavioral, statistical, and rule-based evidence can each catch what the others miss.

5.1 Deep Sequence Analysis Engine

The first component is a deep sequence analysis engine inspired by Transformer-based language models. This engine learns normal event sequences from historical log data and assigns anomaly scores based on deviations from expected behavioral patterns. Unusual login sequences, unexpected authentication flows, credential stuffing attempts, session hijacking patterns, and abnormal user activity produce higher sequence anomaly scores, enabling the detection of attacks that cannot be identified using simple rule-based methods.

5.2 Statistical Anomaly Detection Engine

The second detection engine focuses on statistical anomaly detection using multivariate analysis techniques. Numerical features extracted during feature engineering are analyzed to identify outliers using statistical scoring methods. Rather than relying solely on predefined thresholds, the statistical engine continuously compares current activity with historical behavioral baselines to identify sudden spikes in login failures, abnormal authentication frequencies, impossible travel scenarios, excessive privilege escalation requests, or other significant deviations from normal behavior.

5.3 Cybersecurity Rule Engine

Complementing these AI-driven approaches is a cybersecurity rule engine that implements domain-specific detection logic based on established security heuristics. Rules are designed to identify well-known attack patterns such as repeated failed login attempts indicating brute-force attacks, rapid authentication attempts across multiple accounts suggesting credential stuffing, repeated privilege escalation attempts, and suspicious authentication chains. These rules provide deterministic detection capabilities while also contributing contextual information to the final decision-making process.

6. Hybrid Fusion & Adaptive Thresholding

Rather than relying on a single detection model, AEGIS-SOC employs a hybrid fusion mechanism that combines the outputs of all three detection engines into a unified anomaly score. Each engine contributes to the final decision according to predefined weights, allowing behavioral analysis, statistical evidence, and cybersecurity rules to complement one another.

The resulting anomaly score is evaluated using an adaptive threshold that dynamically adjusts according to the overall behavior of the monitored environment. This adaptive approach significantly reduces false positives

while maintaining high sensitivity to genuine threats. Because multiple independent detection mechanisms must agree before a high-risk alert is generated, analysts receive fewer unnecessary notifications while retaining visibility into sophisticated attack scenarios.

Component	Role in Final Decision
Deep Sequence Engine	Behavioral / sequential deviation signal
Statistical Engine	Baseline deviation & outlier evidence
Cybersecurity Rule Engine	Deterministic, known-pattern signal

7. SIEM Correlation & Threat Intelligence Enrichment

Once anomalous behavior has been detected, the platform enters the SIEM correlation phase. Individual suspicious events are rarely sufficient to represent a complete cyberattack, so AEGIS-SOC correlates multiple related events into coherent attack chains.

Authentication failures, privilege escalation attempts, repeated login requests, and suspicious network activity originating from the same source are grouped together using stateful correlation directives inspired by enterprise SIEM systems. Threat intelligence enrichment further enhances these correlated incidents by checking suspicious IP addresses and indicators against external threat intelligence sources, allowing the platform to prioritize attacks associated with known malicious infrastructure.

8. Risk Scoring & Explainability

Every correlated incident is assigned a dynamic risk score ranging from zero to ten. The risk calculation considers the anomaly confidence generated by the machine learning models, statistical deviation from historical baselines, severity of triggered cybersecurity rules, threat intelligence reputation, and correlation complexity.

Along with the numerical score, the system generates an explainable summary describing exactly why the alert was raised. Instead of presenting analysts with a generic anomaly notification, AEGIS-SOC highlights factors such as repeated authentication failures, unusual login times, unknown devices, suspicious source locations, privilege escalation attempts, and historical behavioral deviations. This explainability improves analyst trust and accelerates incident investigation by providing immediate context behind every generated alert.

Sample Explainable Alert

Field	Example Value
Severity	HIGH
Risk Score	9.6 / 10
Directive	Repeated Authentication Attack Series
Contributing Factors	Repeated auth failures · unfamiliar source IP · known-malicious reputation match

9. Backend Architecture

The backend of the platform is implemented using FastAPI, providing high-performance REST APIs and WebSocket communication for real-time security monitoring. REST endpoints expose key functionalities such as log ingestion, alarm retrieval, KPI monitoring, analytics, user behavior profiles, and model health information, while WebSockets continuously stream live security events and newly generated alerts to connected dashboard clients without requiring page refreshes.

PostgreSQL serves as the primary database for storing processed events, user profiles, alarms, and system metadata, while SQLAlchemy manages database interactions and Alembic supports schema migrations. JWT-based authentication ensures secure access to administrative functionality and analyst operations.

Method	Endpoint	Purpose
GET	/api/kpis	Real-time telemetry: log counts, active threats, critical alarms, MTTD
GET	/api/alarms	SIEM alarms, filterable by status and severity
PUT	/api/alarms/{id}/status	Analyst triage status update (acknowledge / resolve)
GET / POST	/api/logs, /api/logs/ingest	Recent log history / simulate a log through the pipeline
GET	/api/user-profiles	UEBA risk scores and per-user anomaly baselines
GET	/api/analytics	Attack-vector frequency and threat-origin chart data

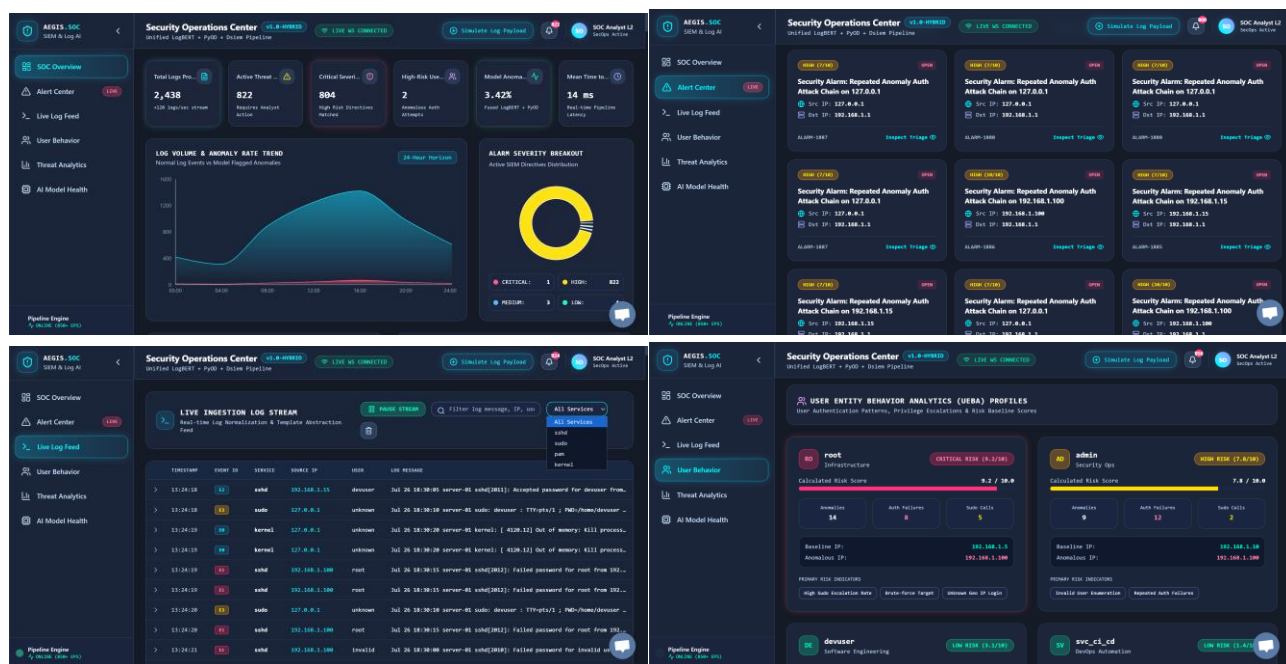
Method	Endpoint	Purpose
GET / POST	/api/model-health[/threshold]	AI model telemetry and live threshold tuning
WS	/ws/live-stream	Broadcasts live log events and new alarms in real time

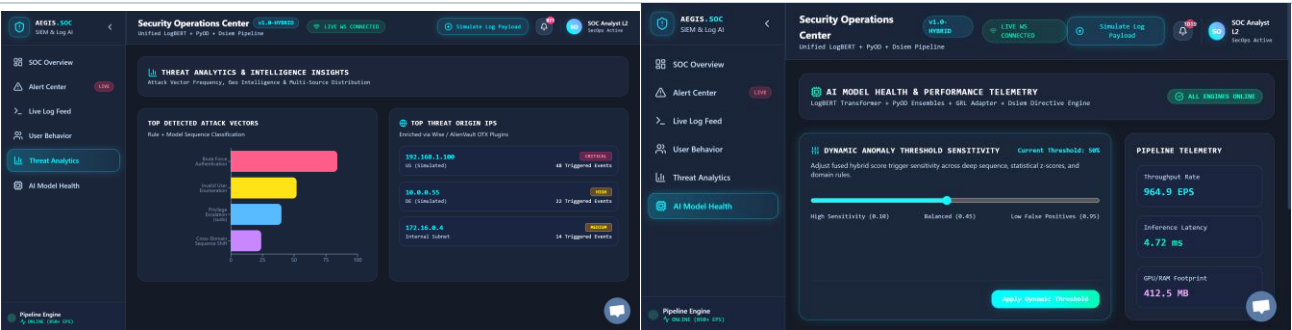
10. Frontend SOC Dashboard

The frontend is developed using React, TypeScript, Tailwind CSS, and Recharts to provide an enterprise-grade Security Operations Center dashboard. The dashboard is divided into multiple specialized modules, including an overview page displaying key performance indicators and threat summaries, a live log feed showing real-time event ingestion, an alert center supporting analyst triage and incident resolution, a User and Entity Behavior Analytics (UEBA) module visualizing behavioral risk scores, a threat analytics section presenting attack trends and threat origins, and a model health dashboard monitoring AI inference latency, throughput, anomaly thresholds, and overall system performance.

Interactive visualizations, real-time WebSocket updates, searchable tables, severity filters, and dynamic charts provide security analysts with comprehensive situational awareness while enabling rapid investigation of suspicious activities.

(Screenshots)





Module	Purpose
Overview	KPIs and threat summaries at a glance
Live Log Feed	Real-time streaming event ingestion
Alert Center	Analyst triage and incident resolution
UEBA	Behavioral risk scores per user/entity
Threat Analytics	Attack trends and threat-origin visualization
Model Health	Inference latency, throughput, live threshold tuning

11. Modular Architecture, Deployment & Testing

The modular architecture of AEGIS-SOC ensures scalability, maintainability, and ease of future enhancement. Each component — including log ingestion, parsing, feature engineering, anomaly detection, score fusion, SIEM correlation, backend services, and frontend visualization — operates as an independent module, allowing new log sources, machine learning models, detection algorithms, or threat intelligence integrations to be incorporated without major architectural changes.

The system is containerized using Docker to simplify deployment across different environments and supports automated testing to ensure reliable operation.

Layer	Technologies
Backend & Infra	Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, JWT auth, WebSockets, Docker

Layer	Technologies
Machine Learning	Transformer-style sequence model, multivariate statistical outlier detection, rule engine
Frontend	React, TypeScript, Tailwind CSS, Recharts

12. Conclusion & Future Scope

Overall, AEGIS-SOC demonstrates how behavioral analytics, machine learning, statistical anomaly detection, and intelligent SIEM correlation can be integrated into a unified cybersecurity platform capable of detecting both known and unknown threats in real time. By reducing false positives, providing explainable alerts, continuously learning organizational behavior, and delivering actionable insights through a modern SOC dashboard, the platform enables faster incident response, improved security operations, and enhanced resilience against increasingly sophisticated cyberattacks.

It serves as a strong foundation for next-generation enterprise SOC environments and can be further extended with additional capabilities as the platform matures:

- MITRE ATT&CK framework mapping for standardized attack classification
- Cloud-native, multi-tenant log monitoring and deployment
- Automated SOAR (Security Orchestration, Automation & Response) workflows
- Large language model-assisted threat investigation and guided hunting
- Predictive cyber threat intelligence and forecasting

